

EARS AND HEARING

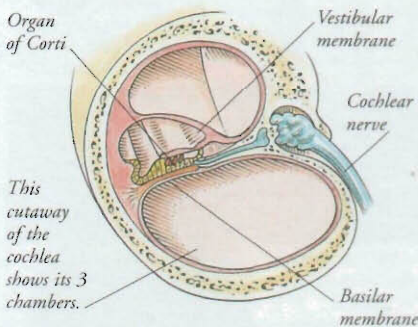
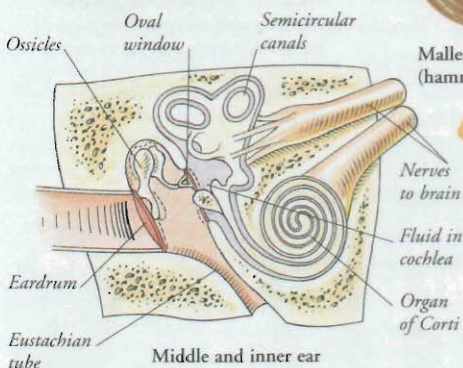


WHEN A BEE BUZZES, a soprano sings, or a jumbo jet takes off, each generates invisible vibrations called sound waves that enter the ears, the

body's organs of hearing. The sound waves travel deep inside the skull to the part of the ear that does the hearing. Here, sound waves are converted into nerve impulses that travel along nerves to the auditory, or hearing, area on each side of the brain. In the brain, the impulses are interpreted as sounds. The ears can pick up a wide range of sounds and, with the eyes, they help us to make sense of our surroundings.

Hearing sounds

Sound waves channelled into the auditory canal cause the eardrum and the ossicles to vibrate. These vibrations travel through the fluid-filled cochlea. Inside the cochlea, sensory hair cells convert the vibrations into nerve impulses. These are carried by the cochlear nerve to the brain.



Cochlea

The cochlea is a long, coiled tube in the inner ear that is filled with fluid. It is divided by two membranes into three chambers that run lengthways. The middle of these three chambers, the cochlear chamber, contains the spiral organ of Corti, which consists of over 20,000 sensory hair cells that send nerve signals to the brain.

Eardrum

The eardrum, or tympanic membrane, is a taut piece of skin that separates the auditory canal from the middle ear. When sound waves hit the eardrum, it vibrates like a drum and transmits its vibrations to the ossicles of the middle ear.



Malleus (hammer)

Incus (anvil)

Stapes (stirrup)

Ossicles

The ossicles are the three smallest bones in the body. The malleus, incus, and stapes connect the eardrum to the cochlea by way of the oval window.

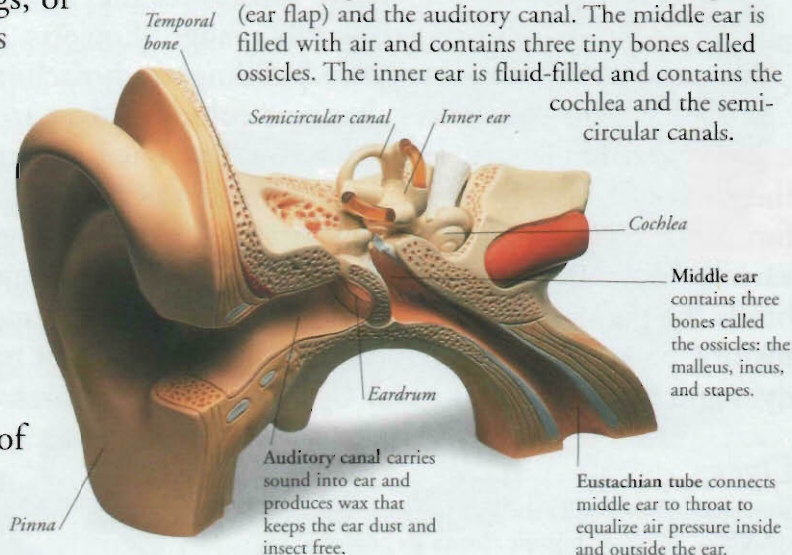
Bartolomeo Eustachio

Italian anatomist Bartolomeo Eustachio (1520–74) studied the detailed anatomy of the ear, as well as other body organs and systems, while he was a professor in Rome. He wrote the first full description of the ears in his book *The Examination of the Organ of Hearing*, published in 1562. Included in this was the first detailed description of the tube that links the middle ear with the throat. This was later named the Eustachian tube.



Anatomy of the ear

Mostly concealed within the skull, the ear is divided into three parts. The outer ear consists of the pinna (ear flap) and the auditory canal. The middle ear is filled with air and contains three tiny bones called ossicles. The inner ear is fluid-filled and contains the cochlea and the semi-circular canals.

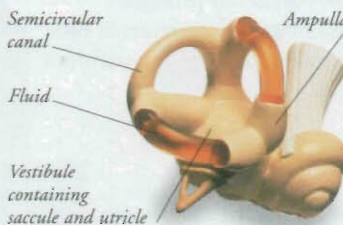
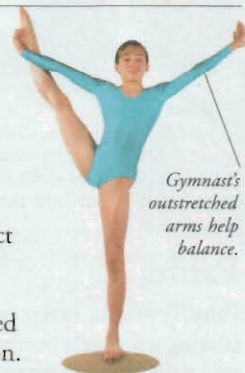


Air pressure

You hear most clearly when the air pressure inside the middle ear is the same as the air pressure outside your body. If the air pressure outside changes suddenly, you may not be able to hear properly. This can happen if you are on a plane that is taking off or landing, or if you are travelling on a fast train.

Balance

Part of the inner ear helps you to balance. Sensors inside the three semicircular canals detect movements made by the head and the rest of the body. Sensors inside two adjoining chambers, the saccule and utricle, detect whether the body is upright, upside-down, or in between. Nerve impulses from the semicircular canals are analysed by the brain to assess the body's position.



Semicircular canals

The three semicircular canals in each ear are filled with fluid. At the base of each canal is a bulge, called an ampulla, which contains sensory hair cells that send impulses to the brain. The three canals are set at 90° to each other, so they can detect movement in any direction.

Hearing ranges

The pitch of a sound depends on the frequency of the sound waves that produced it. High-pitched sounds have a high frequency, and low-pitched sounds have a low frequency. Frequency is measured in units called Hertz (Hz). Our hearing ability decreases as we get older, from 20,000 to 12,000 Hz.



20–20,000 Hz



Bats' ears can hear very high-pitched sound waves called ultrasound.

1,000–120,000 Hz

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